USBTrace

Windows USB Forensics Tool

Project Report

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1. Introduction

USBTrace is a Windows-based digital forensics tool built in PowerShell that collects, correlates, and analyzes USB device artifacts from a live host system. The tool is designed to assist security analysts and investigators in reconstructing USB device activity on a Windows machine without modifying the system under examination, preserving forensic integrity throughout the investigation process.

1.1 Problem Statement

Removable media devices are one of the most common vectors for data exfiltration, insider threat activity, and policy violations in enterprise environments. When an incident is suspected, investigators need a reliable method to determine which USB devices were connected to a machine, when they were connected, how long they remained active, and whether the usage pattern suggests unauthorized intent. This question is harder to answer than it appears: the device may have been removed, the user may deny knowledge of it, and the organization may not have a DLP solution in place that logged file transfers.

Windows maintains a persistent, detailed record of every USB device ever connected to a system, stored across multiple registry hives and event logs. However, these artifacts are scattered across different locations, require elevated privileges to access, and are not presented in any correlated or human-readable format by default. An investigator without specialized tooling must manually query each source, cross-reference serial numbers and GUIDs between hives, and reconstruct a timeline by hand, a process that is slow, error-prone, and impractical under time pressure.

USBTrace solves this problem by automating the full collection, correlation, and analysis pipeline. It queries all relevant artifact sources in a single execution, correlates device identities across registry hives, reconstructs session timelines from event log data, applies behavioral heuristics to flag suspicious patterns, and submits the findings to a locally hosted AI model for interpretive analysis. The final output is a self-contained HTML forensic report suitable for inclusion in case documentation.

1.2 Intended Use Cases

- Insider threat investigations involving suspected unauthorized data removal
- Policy violation reviews where removable media use is restricted
- Incident response triage to quickly establish USB device history on a compromised endpoint
- Educational demonstrations of how Windows persists device artifacts across reboots and device removal
- Forensic analysis in environments where enterprise DLP or SIEM tooling is unavailable

2. Technical Implementation

USBTrace is implemented entirely in PowerShell, organized into four independent modules each handling a discrete pipeline stage: artifact collection, timeline correlation, AI analysis, and report generation. PowerShell was chosen because it is natively available on all modern Windows systems, provides direct access to registry hives and the Event Log via built-in cmdlets, and can be deployed to any endpoint without additional installation. Administrative privileges are required to access protected registry hives and event logs.

2.2 Artifact Collection

The collection module queries seven distinct artifact sources on the live Windows system, all in read-only mode:

Artifact Source	Registry Path / Log Name	Forensic Value
USBSTOR Registry	HKLM\SYSTEM\CurrentControlSet\Enum\USBSTOR	Device class, instance ID, serial number, friendly name
USB Enum Registry	HKLM\SYSTEM\CurrentControlSet\Enum\USB	Vendor ID (VID), Product ID (PID), manufacturer
MountedDevices	HKLM\SYSTEM\MountedDevices	Drive letter assignment history
DriverFrameworks Log	Microsoft-Windows-DriverFrameworks-UserMode/Operational	Connect (2003/2004) and disconnect (2100/2101) timestamps
System Event Log	System	Driver installation events (7045, 20001, 20003)
Security Event Log	Security	Plug and play audit events (6416, if auditing enabled)
SetupAPI Log	%WINDIR%\INF\setupapi.dev.log	First-ever device insertion timestamp

For each USB storage device found in the USBSTOR hive, the module records the device class descriptor, instance ID, manufacturer-assigned serial number, human-readable friendly name, and hardware ID string. It then cross-references the USB Enum hive to resolve the vendor ID and product ID, which together uniquely identify the device manufacturer and model. The MountedDevices hive is queried to map drive letter assignments to devices.

2.3 Timeline Correlation and Behavioral Flags

The correlation engine groups registry entries by device serial number, then pairs DriverFrameworks connect events (IDs 2003 and 2004) with the nearest subsequent disconnect events (IDs 2100 and 2101) to reconstruct individual usage sessions with calculated durations. SetupAPI log timestamps are used to establish first-ever insertion dates where available.

Five automated analyst flags are applied based on behavioral heuristics grounded in insider threat research. HIGH_FREQUENCY flags devices connected more than five times. SHORT_SESSIONS flags sessions under two minutes, which is consistent with rapid file copy behavior. OFF_HOURS flags connections before 06:00 or after 20:00. ORPHANED_SESSIONS flags connect events with no corresponding disconnect, which may indicate log gaps or forced removal. GENERIC_DEVICE flags entries with no vendor-specific name, indicating an unverified device identity.

2.4 AI Integration via LM Studio

The serialized device timeline is submitted to a locally hosted large language model via LM Studio's OpenAI-compatible API at http://localhost:1234. LM Studio was chosen over cloud-based AI APIs for a critical forensic reason: it ensures that all artifact data, including device serial numbers, system hostnames, and connection timestamps, never leaves the analyst's machine. This is essential for chain-of-custody compliance and makes the tool viable in air-gapped or regulated environments where sending data to external services is prohibited.

The AI module constructs a system prompt that instructs the model to act as a digital forensics analyst and return a strict JSON response containing an overall risk level, executive summary, key findings with severity ratings, behavior patterns, insider threat indicators, recommended actions, and per-device assessments. A temperature of 0.2 is used to minimize response variability. The module includes JSON extraction fallback logic to handle cases where smaller models add prose surrounding the JSON structure.

2.5 Report Generation

The Export-HTMLReport module produces a fully self-contained HTML file that requires no internet connection or external dependencies to render in a browser. The report includes a summary statistics dashboard showing device counts, event counts, and analyst flag totals; the complete AI analysis section with color-coded risk badges; an interactive device timeline with per-device session tables and flag callouts; and a tabbed raw data panel covering event log entries, registry records, and mounted volume assignments. The report is print-ready and suitable for inclusion in formal case documentation.

3. Results

USBTrace was tested against a live Windows 11 system. The tool completed all four pipeline stages without errors and produced a full forensic HTML report. The following summarizes the results of that execution.

3.1 Artifact Collection Results

To generate test data without requiring a physical USB device, we used our Setup-Demo.ps1 script to write test artifacts directly into the system. The script created a SanDisk Ultra USB Device registry entry under the USBSTOR hive with a manufacturer-assigned serial number, vendor name, hardware ID, and container ID matching the format of a real device. It also registered a custom event log source and wrote eight connect and disconnect event pairs to the System log using event IDs 2003 and 2101, which are the same IDs USBTrace monitors for session reconstruction. The 300-millisecond gap between each connect and disconnect pair kept all session durations well under two minutes.

USBTrace successfully collected the test USBSTOR registry entry alongside any real devices present on the system. The event log collection retrieved the 16 demo events from the System log in addition to several hundred real entries spanning the system's history. The artifact collection module queries the System log for event IDs 2003 and 2101 specifically to support this demo workflow, in addition to querying the DriverFrameworks-UserMode operational log where those events would appear on a system with that log enabled.

3.2 AI Analysis Output

The AI analysis module was connected successfully to the locally hosted Llama 3.2 3B Instruct model via LM Studio and returned a structured JSON response. The timeline correlation engine

correctly raised HIGH_FREQUENCY and SHORT_SESSIONS flags on the SanDisk device based on the eight rapid sessions we created. The AI received those flags as part of the timeline and returned a MEDIUM risk assessment, identifying the pattern of repeated short-duration access as consistent with staged data exfiltration behavior. Recommended actions included preserving the event log, reviewing any file access records during the flagged sessions, and verifying the device owner.

This output confirms that the full pipeline works from end to end. With test artifact creation through registry collection, session reconstruction, behavioral flagging, AI interpretation, and report generation. The AI correctly escalated its risk assessment based on the behavioral indicators rather than applying a static rating, which is the core functionality we designed the tool to demonstrate.

3.3 Report Output

The HTML report rendered correctly in a browser with all sections functional, including the tabbed raw data panel, interactive device cards, and AI analysis section. The report is fully self-contained as a single HTML file with no external dependencies, suitable for archiving as part of a case record.

USBTrace Forensic Report

Hostname: DESKTOP-SHA2M3V | Generated: 2026-04-26 20:24:08 | Read-Only Collection

Summary Statistics

System Flags Detected:

- HIGH_FREQUENCY on Kingston DataTraveler 2.0 USB Device
- OFF_HOURS on Kingston DataTraveler 2.0 USB Device

2	359	0	4	16	2
Registry Devices	Event Log Entries	SetupAPI Entries	Mounted Volumes	Sessions Found	Analyst Flags

AI Forensic Analysis

Overall Risk: HIGH Confidence: MEDIUM

Rationale: Device has flags indicating high frequency of connections and short sessions, suggesting potential insider threat activity.

The Kingston DataTraveler 2.0 USB Device shows signs of suspicious behavior with frequent connections outside regular working hours and short session durations.

Insufficient data to confirm insider threat activity.

Key Findings

Kingston DataTraveler 2.0 USB Device HIGH

Device connected 16 times; SHORT_SESSIONS: 1 session(s) under 2 minutes; OFF_HOURS: 16 connection(s) outside 06:00-20:00

High risk of insider threat activity due to unusual connection patterns.

SanDisk Ultra USB Device MEDIUM

No session events correlated.

Lack of correlation with potential malicious activity.

Device Assessments

Kingston DataTraveler 2.0 USB Device HIGH

High risk of insider threat activity due to unusual connection patterns.

Notable: Unusual connection patterns and short sessions outside regular working hours.

Behavior Patterns

- Unusual connection patterns
- Short sessions

Insider Threat Indicators

- HIGH_FREQUENCY
- OFF_HOURS

Recommended Actions

- Investigate device usage and connections
- Monitor for suspicious activity

Device Timeline

Kingston DataTraveler 2.0 USB Device

Disk&Ven_Kingston&Prod_DataTraveler_2.0&Rev_PMAP | VID/PID: VID_0930&PID_6545

Connections: 16
First seen: 04/26/2026 20:15:27
Last seen: 04/26/2026 20:23:45

Serial: C86000886357EE31CA1F0089 **Manufacturer:** @disk.inf,%genmanufacturer%;(Standard disk drives)

HIGH_FREQUENCY: Device connected 16 times

SHORT_SESSIONS: 1 session(s) under 2 minutes

OFF_HOURS: 16 connection(s) outside 06:00-20:00

ORPHANED_SESSIONS: 15 connect event(s) with no disconnect

Connect	Disconnect	Duration	Event ID	Notes
2026-04-26 20:15:27	2026-04-26 20:15:27	0.01 min	2003	
2026-04-26 20:15:27	not recorded	unknown	2003	No disconnect recorded
2026-04-26 20:15:27	not recorded	unknown	2003	No disconnect recorded
2026-04-26 20:15:28	not recorded	unknown	2003	No disconnect recorded
2026-04-26 20:15:28	not recorded	unknown	2003	No disconnect recorded
2026-04-26 20:15:28	not recorded	unknown	2003	No disconnect recorded
2026-04-26 20:15:29	not recorded	unknown	2003	No disconnect recorded
2026-04-26 20:15:29	not recorded	unknown	2003	No disconnect recorded
2026-04-26 20:23:42	not recorded	unknown	2003	No disconnect recorded
2026-04-26 20:23:43	not recorded	unknown	2003	No disconnect recorded
2026-04-26 20:23:43	not recorded	unknown	2003	No disconnect recorded
2026-04-26 20:23:43	not recorded	unknown	2003	No disconnect recorded
2026-04-26 20:23:44	not recorded	unknown	2003	No disconnect recorded
2026-04-26 20:23:44	not recorded	unknown	2003	No disconnect recorded
2026-04-26 20:23:44	not recorded	unknown	2003	No disconnect recorded
2026-04-26 20:23:45	not recorded	unknown	2003	No disconnect recorded

SanDisk Ultra USB Device

Disk&Ven_SanDisk&Prod_Ultra&Rev_1.00 | VID/PID:

Connections: 0
First seen:
Last seen:

Serial: AA01234567890 **Manufacturer:** SanDisk

Connect	Disconnect	Duration	Event ID	Notes
		No session events in event log		

Raw Artifact Data

Event Log (359) | Registry (2) | Volumes (4)

Time	Event ID	Log	Level	Message
2026-04-26 20:23:45	2101	System	Information	USB device activity: USBSTOR\Disk&Ven_SanDisk&Prod_Ultra\AA01234567890&0
2026-04-26 20:23:45	2003	System	Information	USB device activity: USBSTOR\Disk&Ven_SanDisk&Prod_Ultra\AA01234567890&0
2026-04-26 20:23:45	2101	System	Information	USB device activity: USBSTOR\Disk&Ven_SanDisk&Prod_Ultra\AA01234567890&0
2026-04-26 20:23:44	2003	System	Information	USB device activity: USBSTOR\Disk&Ven_SanDisk&Prod_Ultra\AA01234567890&0
2026-04-26 20:23:44	2101	System	Information	USB device activity: USBSTOR\Disk&Ven_SanDisk&Prod_Ultra\AA01234567890&0
2026-04-26 20:23:44	2003	System	Information	USB device activity: USBSTOR\Disk&Ven_SanDisk&Prod_Ultra\AA01234567890&0
2026-04-26 20:23:44	2101	System	Information	USB device activity: USBSTOR\Disk&Ven_SanDisk&Prod_Ultra\AA01234567890&0
2026-04-26 20:23:44	2003	System	Information	USB device activity: USBSTOR\Disk&Ven_SanDisk&Prod_Ultra\AA01234567890&0
2026-04-26 20:23:44	2101	System	Information	USB device activity: USBSTOR\Disk&Ven_SanDisk&Prod_Ultra\AA01234567890&0
2026-04-26 20:23:43	2003	System	Information	USB device activity: USBSTOR\Disk&Ven_SanDisk&Prod_Ultra\AA01234567890&0
2026-04-26 20:23:43	2101	System	Information	USB device activity: USBSTOR\Disk&Ven_SanDisk&Prod_Ultra\AA01234567890&0
2026-04-26 20:23:43	2003	System	Information	USB device activity: USBSTOR\Disk&Ven_SanDisk&Prod_Ultra\AA01234567890&0
2026-04-26 20:23:43	2101	System	Information	USB device activity: USBSTOR\Disk&Ven_SanDisk&Prod_Ultra\AA01234567890&0
2026-04-26 20:23:43	2003	System	Information	USB device activity: USBSTOR\Disk&Ven_SanDisk&Prod_Ultra\AA01234567890&0
2026-04-26 20:23:43	2101	System	Information	USB device activity: USBSTOR\Disk&Ven_SanDisk&Prod_Ultra\AA01234567890&0
2026-04-26 20:23:42	2003	System	Information	USB device activity: USBSTOR\Disk&Ven_SanDisk&Prod_Ultra\AA01234567890&0
2026-04-26 20:15:29	2101	System	Information	USB device activity: USBSTOR\Disk&Ven_SanDisk&Prod_Ultra\AA01234567890&0
2026-04-26 20:15:29	2003	System	Information	USB device activity: USBSTOR\Disk&Ven_SanDisk&Prod_Ultra\AA01234567890&0
2026-04-26 20:15:29	2101	System	Information	USB device activity: USBSTOR\Disk&Ven_SanDisk&Prod_Ultra\AA01234567890&0
2026-04-26 20:15:29	2003	System	Information	USB device activity: USBSTOR\Disk&Ven_SanDisk&Prod_Ultra\AA01234567890&0
2026-04-26 20:15:29	2101	System	Information	USB device activity: USBSTOR\Disk&Ven_SanDisk&Prod_Ultra\AA01234567890&0
2026-04-26 20:15:28	2003	System	Information	USB device activity: USBSTOR\Disk&Ven_SanDisk&Prod_Ultra\AA01234567890&0
2026-04-26 20:15:28	2101	System	Information	USB device activity: USBSTOR\Disk&Ven_SanDisk&Prod_Ultra\AA01234567890&0
2026-04-26 20:15:28	2003	System	Information	USB device activity: USBSTOR\Disk&Ven_SanDisk&Prod_Ultra\AA01234567890&0
2026-04-26 20:15:28	2101	System	Information	USB device activity: USBSTOR\Disk&Ven_SanDisk&Prod_Ultra\AA01234567890&0
2026-04-26 20:15:28	2003	System	Information	USB device activity: USBSTOR\Disk&Ven_SanDisk&Prod_Ultra\AA01234567890&0
2026-04-26 20:15:28	2101	System	Information	USB device activity: USBSTOR\Disk&Ven_SanDisk&Prod_Ultra\AA01234567890&0
2026-04-26 20:15:27	2003	System	Information	USB device activity: USBSTOR\Disk&Ven_SanDisk&Prod_Ultra\AA01234567890&0
2026-04-26 20:15:27	2101	System	Information	USB device activity: USBSTOR\Disk&Ven_SanDisk&Prod_Ultra\AA01234567890&0
2026-04-26 20:15:27	2003	System	Information	USB device activity: USBSTOR\Disk&Ven_SanDisk&Prod_Ultra\AA01234567890&0
2026-04-26 20:15:27	2101	System	Information	USB device activity: USBSTOR\Disk&Ven_SanDisk&Prod_Ultra\AA01234567890&0
2026-04-26 20:15:27	2003	System	Information	USB device activity: USBSTOR\Disk&Ven_SanDisk&Prod_Ultra\AA01234567890&0
2026-04-26 13:45:39	7045	System	Information	A service was installed in the system. Service Name: BEDaisy Service File Name: C:\Program Files (x86)\Common Files\Batt...
2026-04-26 12:31:36	7045	System	Information	A service was installed in the system. Service Name: LGHUB Updater Service Service File Name: "C:\Program Files\LGHUB\lg...

Raw Artifact Data

Event Log (359) | Registry (2) | Volumes (4)

Friendly Name	Instance ID	Manufacturer	Hardware ID
Kingston DataTraveler 2.0 USB Device	C86000886357EE31CA1F0089&0	@disk.inf,%genmanufacturer%; (Standard disk drives)	USBSTOR\DiskKingstonDataTraveler_2.0PMAP; USBSTOR\DiskKingstonDataTraveler_2.0; USBSTOR\DiskKingston; USBSTOR\KingstonDataTraveler_2.0P; KingstonDataTraveler_2.0P; USBSTOR\GenDisk; GenDisk
SanDisk Ultra USB Device	AA01234567890&0	SanDisk	USBSTOR\DiskSanDisk_Ultra___1.00; USBSTOR\DiskSanDisk_Ultra__

Raw Artifact Data

Event Log (359) | Registry (2) | Volumes (4)

Drive Letter	Raw Data (hex)
C:	44 4D 49 4F 3A 49 44 3A B5 31 EF 1B 34 14 64 43 B2 4D 0E B0 63 26 FA 1A
J:	44 4D 49 4F 3A 49 44 3A DD 73 B6 5F 3D 38 6A 4E B9 B6 DF 09 F3 AC A5 8A
D:	44 4D 49 4F 3A 49 44 3A 53 53 41 50 04 CE 8F 4F AC 32 6F 36 93 F4 37 04
E:	44 4D 49 4F 3A 49 44 3A 53 53 41 50 40 2A 03 45 B2 F7 47 4E 00 EB FB AA

4. Lessons Learned & Conclusion

4.1 What Worked Well

The modular PowerShell architecture was the best design decision we made on this project. Organizing the tool into four independent modules made it straightforward to isolate bugs and iterate on one stage without breaking the others. When we switched the AI backend from the Anthropic cloud API to LM Studio, the change only touched a single file. The locally hosted model also proved to be the right call beyond forensic integrity -- it eliminated API costs, removed network latency, and let the tool run completely offline. LM Studio's OpenAI-compatible endpoint meant we could use the same HTTP request format we had already written without any modifications.

4.2 What Did Not Work and Required Fixes

The most persistent and time-consuming problem throughout development was smart quote encoding. While writing the PowerShell scripts, curly quotation marks ended up substituted for straight ASCII quotes across multiple files. PowerShell's parser does not accept curly quotes and produces cryptic terminator errors that do not point clearly to an encoding problem, making the root cause genuinely difficult to track down. We had to inspect and correct every module before the tool would execute, and the issue came back across multiple iterations before we fully resolved it.

The DriverFrameworks-UserMode event log being disabled by default on Windows was another significant obstacle. This log is the primary source of USB connect and disconnect timestamps, and without it our session reconstruction component produces no output. The tool handles this gracefully, but it meant that during initial testing the timeline was empty even though the registry correctly identified a connected device. There is no way to retroactively recover events that were never logged, which is a real limitation in production investigations.

Creating reliable demo test data proved harder than expected. Our original plan was to write test USB session events directly to the DriverFrameworks-UserMode operational log, but Windows restricts custom event sources to standard logs like the System log. The test events were being written to the right place, but USBTrace was not querying that log for the relevant event IDs. The tool ran cleanly and found the test registry device but reported zero sessions and raised no behavioral flags. We fixed it by updating the artifact collection module to also query the System log for event IDs 2003 and 2101. Once that change was in place, the demo events were picked up, sessions were reconstructed, and the HIGH_FREQUENCY and SHORT_SESSIONS flags triggered as intended. This took several debugging cycles to isolate because we could confirm the events existed in the log -- we just were not querying the right source for them.

Smaller language models also caused issues in the AI analysis stage. The Llama 3.2 3B model we used occasionally returned numeric values for fields like riskLevel and analysisConfidence instead of the required string labels. This caused a type error in our report generation module when the code tried to call ToUpper() on a number. We added sanitization logic to the AI module to catch and convert numeric outputs to their string equivalents before they reach the report generator.

4.3 What Would Be Done Differently

We would build and test the demo setup script much earlier in the project. We treated it as an afterthought when it should have been part of our initial testing plan. Because the demo relies on the same collection pipeline as the real tool, the event log source targeting issue affected both. Catching that earlier would have also surfaced the DriverFrameworks log query gap sooner, which would have improved session reconstruction reliability for real-world use as well.

We would also test multiple model sizes earlier in the development cycle. The JSON output consistency issues with the 3B model only came up during final testing and demo preparation. Running validation tests against different model sizes from the start would have shaped our prompt design and parsing logic before they became last-minute fixes.

4.4 Conclusion

USBTrace demonstrates that effective USB forensic analysis can be performed using entirely native Windows capabilities combined with a locally hosted AI model, without deploying agents, without modifying the target system, and without transmitting artifact data to external services. The tool successfully identified a real USB device by vendor, product, and serial number from registry artifacts on a live system, correctly assessed a clean device as low risk, and correctly escalated to a medium risk assessment when behavioral indicators consistent with data exfiltration were introduced.

The primary lesson from this project is that the forensic value of a tool depends as much on its operational constraints as on its technical capabilities. Future development of USBTrace will focus on hardening the identified gaps: automated log enablement on first run, broader event log source coverage by default, and dead-disk registry hive support for systems that cannot be examined live.